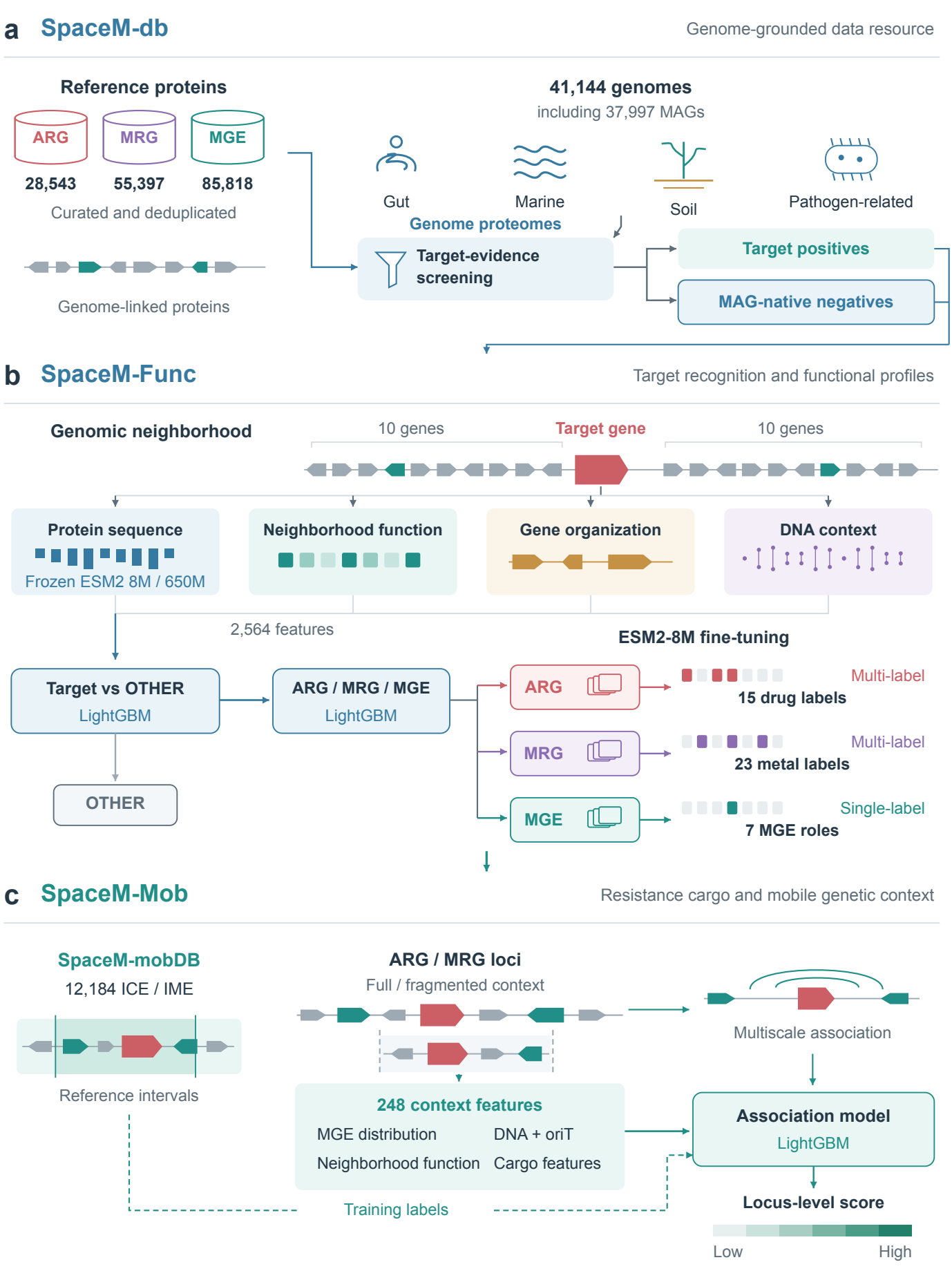

a SpaceM-db
Genome-grounded data resource
Reference proteins
ARG
MRG
MGE
28,543
55,397
85,818
Curated and deduplicated
Genome-linked proteins
41,144 genomes
including 37,997 MAGs
Gut
Marine
Soil
Pathogen-related
Genome proteomes
Target-evidence screening
Target positives
MAG-native negatives
b SpaceM-Func
Target recognition and functional profiles
Genomic neighborhood
10 genes
Target gene
10 genes
Protein sequence
Frozen ESM2 8M / 650M
Neighborhood function
Gene organization
DNA context
2,564 features
ESM2-8M fine-tuning
Target vs OTHER
LightGBM
ARG / MRG / MGE
LightGBM
OTHER
ARG
Multi-label
15 drug labels
MRG
Multi-label
23 metal labels
MGE
Single-label
7 MGE roles
c SpaceM-Mob
Resistance cargo and mobile genetic context
SpaceM-mobDB
12,184 ICE / IME
Reference intervals
ARG / MRG loci
Full / fragmented context
Multiscale association
248 context features
MGE distribution
DNA + oriT
Neighborhood function
Cargo features
Association model
LightGBM
Training labels
Locus-level score
Low
High